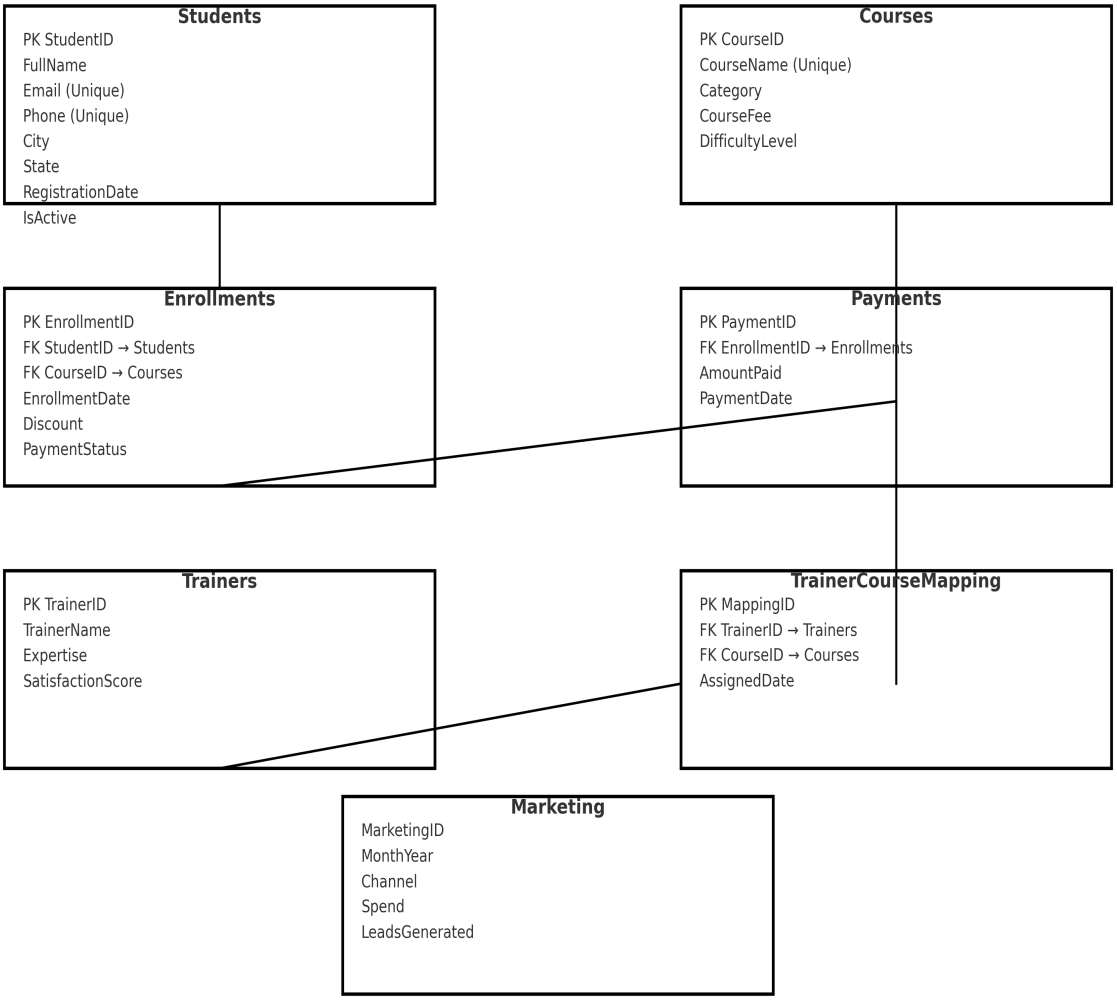
LearnPro EdTech SQL PROJECT – DETAILED ER DIAGRAM & DESCRIPTION



ABOUT THIS ER DIAGRAM

This Entity–Relationship Diagram (ERD) represents the complete relational database structure used in the LearnPro EdTech SQL Project. It defines all entities, attributes, primary keys, foreign keys, and their relational connections that support the analytics-driven EdTech platform.

1. STUDENTS TABLE

Stores information of every student registered in the platform.

Includes:

• PK StudentID

• FullName

• Email (Unique)

• Phone (Unique)

• City

• State

• RegistrationDate

• IsActive

2. COURSES TABLE

Contains details of all courses offered by LearnPro EdTech.

Includes:

• PK CourseID

• CourseName (Unique)

• Category

• CourseFee

• DifficultyLevel

3. ENROLLMENTS TABLE

Tracks every course enrollment by students.

Includes:

• PK EnrollmentID

• FK StudentID → Students

• FK CourseID → Courses

• EnrollmentDate

• Discount

• PaymentStatus

4. PAYMENTS TABLE

Logs all transactions associated with student enrollments.

Includes:

• PK PaymentID

• FK EnrollmentID → Enrollments

• AmountPaid

• PaymentDate

5. TRAINERS TABLE

Stores trainer details and expertise profiles.

Includes:

• PK TrainerID

• TrainerName

• Expertise

• SatisfactionScore

6. TRAINER–COURSE MAPPING TABLE

Represents the many-to-many relationship between trainers and courses.

Includes:

• PK MappingID

• FK TrainerID → Trainers

• FK CourseID → Courses

• AssignedDate

7. MARKETING TABLE

Captures monthly marketing spend and lead generation metrics.

Includes:

• MarketingID

• MonthYear

• Channel

• Spend

• LeadsGenerated

RELATIONSHIPS REPRESENTED

• One Student → Many Enrollments

• One Course → Many Enrollments

• One Enrollment → One Payment

• One Trainer → Many TrainerCourseMappings

• One Course → Many TrainerCourseMappings

• Marketing table is independent (analytics only)

This ERD forms the backbone of all SQL tasks: DDL, joins, analytics, window functions, procedures, triggers, and reporting for LearnPro EdTech.